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On the Utility of Learning about Humans for Human-AI Coordination

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Abstract

While we would like agents that can coordinate with humans, current algorithms such as self-play and population-based training create agents that can coordinate with *themselves*. Agents that assume their partner to be optimal or similar to them can converge to coordination protocols that fail to understand and be understood by humans. To demonstrate this, we introduce a simple environment that requires challenging coordination, based on the popular game *Overcooked*, and learn a simple model that mimics human play. We evaluate the performance of agents trained via self-play and population-based training. These agents perform very well when paired with themselves, but when paired with our human model, they are significantly worse than agents designed to play with the human model. An experiment with a planning algorithm yields the same conclusion, though only when the human-aware planner is given the exact human model that it is playing with. A user study with real humans shows this pattern as well, though less strongly. Qualitatively, we find that the gains come from having the agent *adapt* to the human's gameplay. Given this result, we suggest several approaches for designing agents that learn about humans in order to better coordinate with them. Code is available at https://github.com/HumanCompatibleAI/overcooked_ai.

1 Introduction

An increasingly effective way to tackle two-player games is to train an agent to play with a set of other AI agents, often past versions of itself. This powerful approach has resulted in impressive performance against human experts in games like Go [33], Quake [20], Dota [29], and Starcraft [34].

Since the AI agents never encounter humans during training, when evaluated against human experts, they are undergoing a distributional shift. Why doesn't this cause the agents to fail? We hypothesize that it is because of the *competitive nature* of these games. Consider the canonical case of a two-player zero-sum game, as shown in Figure 1 (left). When humans play the minimizer role but take a branch in the search tree that is suboptimal, this only *increases* the maximizer's score.

However, not all settings are competitive. Arguably, one of the main goals of AI is to generate agents that *collaborate*, rather than compete, with humans. We would like agents that help people with the tasks they want to achieve, augmenting their capabilities [10, 6]. Looking at recent results, it is tempting to think that self-play-like methods extend nicely to collaboration: AI-human teams performed well in Dota [28] and Capture the Flag [20]. However, in these games, the advantage may come from the AI system's individual ability, rather than from *coordination* with humans. We claim that in general, collaboration is fundamentally different from competition, and will require us to go beyond self-play to explicitly account for *human* behavior.

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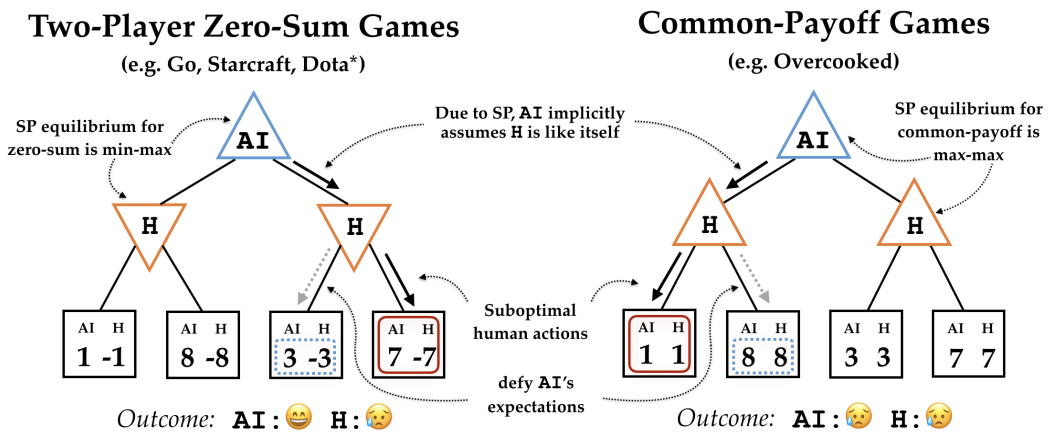

Figure 1: The impact of incorrect expectations of optimality. **Left:** In a competitive game, the agent plans for the worst case. **AI** expects that if it goes left, **H** will go left. So, it goes right where it expects to get 3 reward (since **H** would go left). When **H** suboptimally goes right, **AI** gets 7 reward: *more* than it expected. **Right:** In a collaborative game, **AI** expects **H** to coordinate with it to choose the best option, and so it goes left to obtain the 8 reward. However, when **H** suboptimally goes left, **AI** only gets 1 reward: the worst possible outcome!

Consider the canonical case of a common-payoff game, shown in Figure 1 (right): since both agents are maximizers, a mistake on the human's side is no longer an advantage, but an actual problem, *especially* if the agent did not anticipate it. Further, agents that are allowed to co-train might converge onto opaque coordination strategies. For instance, agents trained to play the collaborative game Hanabi learned to use the hint "red" or "yellow" to indicate that the newest card is playable, which no human would immediately understand [12]. When such an agent is paired with a human, it will execute the opaque policy, which may fail spectacularly when the human doesn't play their part.

We thus hypothesize that in true collaborative scenarios agents trained to play well with other AI agents will perform much more poorly when paired with humans. We further hypothesize that incorporating human data or models into the training process will lead to significant improvements.

To test this hypothesis, we introduce a simple environment based on the popular game *Overcooked* [13], which is specifically designed to be challenging for humans to coordinate in (Figure 2).We use this environment to compare agents trained with themselves to agents trained with a learned human model. For the former, we consider self-play [33], population-based training [19], and coupled planning with replanning. For the latter, we collect human-human data and train a behavior cloning human model; we then train a reinforcement learning and a planning agent to collaborate well with this model. We evaluate how well these agents collaborate with a held-out "simulated" human model (henceforth the "proxy human"), trained on a different data set, as well as in a user study.

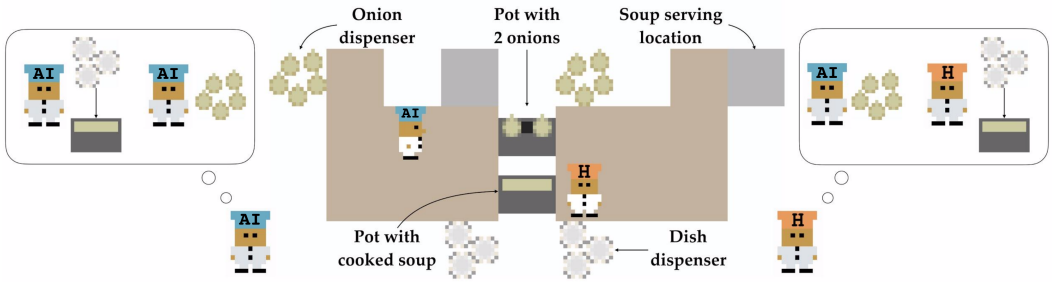

Figure 2: Our Overcooked environment. The goal is to place three onions in a pot (dark grey), take out the resulting soup on a plate (white) and deliver it (light grey), as many times as possible within the time limit. **H**, the human, is close to a dish dispenser and a cooked soup, and **AI**, the agent, is facing a pot that is not yet full. The optimal strategy is for **H** to put an onion in the partially full pot, and for **AI** to put the existing soup in a dish and deliver it. This is due to the layout structure, that makes **H** have an advantage in placing onions in pots, and **AI** have an advantage in delivering soups. However, we can guess that **H** plans to pick up a plate to deliver the soup. If **AI** nonetheless expects **H** to be optimal, it will expect **H** to turn around to get the onion, and will continue moving towards its own dish dispenser, leading to a *coordination failure*.

We find that the agents which did not leverage human data in training perform very well with themselves, and drastically worse when paired with the proxy human. This is not explained only by human suboptimality, because the agent also significantly underperforms a "gold standard" agent that has access to the proxy human. The agent trained with the behavior-cloned human model is drastically better, showing the benefit of having a relatively good human model. We found the same trends even when we paired these agents with real humans, for whom our model has much poorer predictive power but nonetheless helps the agent be a better collaborator. We also experimented with using behavior cloning directly for the agent's policy, and found that it also outperforms self-play-like methods, but still underperforms relative to methods that that leverage planning (with respect to the actual human model) or reinforcement learning with a proxy human model.

Overall, we learned a few important lessons in this work. First, our results showcase the importance of accounting for real human behavior during training: even using a behavior cloning model prone to failures of distributional shift seems better than treating the human as if they were optimal or similar to the agent. Second, leveraging planning or reinforcement learning to maximize the collaborative reward, again even when using such a simple human model, seems to already be better than vanilla imitation. These results are a cautionary tale against relying on self-play or vanilla imitation for collaboration, and advocate for methods that leverage models of human behavior, actively improve them, or even use them as part of a population to be trained with.

2 Related Work

Human-robot interaction (HRI). The field of human robot interaction has already embraced our main point that we shouldn't model the human as optimal. Much work focuses on achieving collaboration by planning and learning with (non-optimal) models of human behavior [26, 21, 31], as well as on specific properties of robot behavior that aid collaboration [2, 14, 9]. However, to our knowledge, ours is the first work to analyze the optimal human assumption in the context of deep reinforcement learning, and to test potential solutions such as population-based training (PBT).

Choudhury et al. [7] is particularly related to our work. It evaluates whether it is useful to learn a human model using deep learning, compared to a more structured "theory of mind" human model. We are instead evaluating how useful it is to have a human model *at all*.

Multiagent reinforcement learning. Deep reinforcement learning has also been applied to multiagent settings, in which multiple agents take actions in a potentially non-competitive game [24, 15]. Some work tries to teach agents collaborative behaviors [22, 18] in environments where rewards are *not* shared across agents. The Bayesian Action Decoder [12] learns communicative policies that allow two agents to collaborate, and has been applied to the cooperative card game Hanabi. However, most multiagent RL research focuses on AI-AI interaction, rather than the human-AI setting.

Lerer and Peysakhovich [23] starts from the same observation that self-play will perform badly in general-sum games, and aims to do better given some data from agents that will be evaluated on at test time (analogous to our human data). However, they assume that the test time agents have settled into an equilibrium that their agent only needs to replicate, and so they train their agent with Observational Self-Play (OSP): a combination of imitation learning and MARL. In contrast, we allow for the case where humans do not play an equilibrium strategy (because they are suboptimal), and so we only use imitation learning to create human models, and train our agent using pure RL.

Imitation learning. Imitation learning [1, 16] aims to train agents that mimic the policies of a *demonstrator*. In our work, we use behavior cloning [30] to imitate demonstrations collected from humans, in order to learn human models to collaborate with. However, the main focus of our work is in the design of agents that can collaborate with these models, and not the models themselves.

3 Preliminaries

Multi-agent MDP. A multiagent Markov decision process [5] is defined by a tuple $\langle S, \alpha, \{A_{i \in \alpha}\}, \mathcal{T}, R \rangle$. S is a finite set of states, and $R : S \to \mathbb{R}$ is a real-valued reward function. α is a finite set of agents; A_i is the finite set of actions available to agent i. $\mathcal{T} : S \times A_1 \times \cdots \times A_n \times S \to [0, 1]$ is a transition function that determines the next state given *all* of the agents' actions.